

Jingtao Zhang

CONTACT zhangjingtao1101@gmail.com
INFORMATION

EDUCATION **Virginia Tech**, Blacksburg, USA

Ph.D., Operations Research, 8/16/2021–8/4/2025

- Courses taken: Optimization, Machine Learning for System Intelligence, Dynamic Programming, Advanced Mathematical Programming, Random Process, Measure and Probability, Statistical Inference, Surrogate Modeling

Shanghai Jiao Tong University, Shanghai, China

M.E., Logistics Engineering, 9/1/2017–3/30/2020

- Courses taken: Stochastic Models, Transportation and Distribution, Sustainable Supply Chain Management, Production and Operations Management
- GPA Rank 1/13

Dalian Maritime University, Liaoning, China

B.E., Logistics Engineering, 9/1/2013–6/30/2017

- Courses taken: Logistics Information Management, Systems Engineering, Techniques and Method of Forecasting, Simulation Technology of Logistics System
- GPA Rank 1/67

RESEARCH Design and analysis of stochastic simulation experiment, uncertainty quantification,
INTERESTS simulation optimization, active learning, sensitivity analysis

WORKING **Research Scientist** 7/14/2025 – Now
EXPERIENCE *Supply Chain Optimization Technologies (SCOT)*

Job description: *I develop reinforcement learning (RL) algorithms for supply chain optimization, focusing on decision-making related to procurement and inventory placement. In addition, I design and analyze A/B tests to evaluate the deployment and performance of RL algorithms in operational supply chain settings. I also develop machine learningbased surrogate models to accelerate large-scale supply chain simulation systems.*

Research Scientist Intern at Amazon 5/20/2024 – 8/23/2024
Supply Chain Optimization Technologies (SCOT)

Job description: *I developed an uncertainty quantification framework to predict quantiles and construct prediction intervals for critical supply chain metrics, achieving a nearly 10% improvement in prediction accuracy. Additionally, I enhanced the framework to enable prediction attribution using Shapley values.*

Research Scientist Intern at Amazon 5/31/2023 – 8/18/2023
Supply Chain Optimization Technologies (SCOT)

Job description: *I developed an advanced simulation module for modeling order quantities in the supply chain, delivering significantly improved accuracy and substantially faster simulation speeds compared to the existing production system. The module was implemented and packaged as a core component in Python.*

- PUBLICATIONS
- J. Zhang and X. Chen, “Sobol’ Index Estimation via Nested Simulation,” *submitted*.
 - J. Zhang and X. Chen, “Global Sensitivity Analysis of Stochastic Models with Temporal Dynamics,” *under major revision*.
 - J. Zhang and X. Chen, “A Model-Free, Partition-Based Approach to Estimating Sobol Indices from Existing Datasets,” *Proceedings of the 2025 Winter Simulation Conference*, 2025.
 - J. Zhang and X. Chen, “A Multilevel Monte Carlo Metamodeling Approach to Variance Function Estimation,” *SIAM/ASA Journal on Uncertainty Quantification*, vol. 13, no. 3, pp. 9801027, 2025,.
 - X. Chen, Y. Zhang, G. Xie, and J. Zhang, “A Uniform Error Bound for Stochastic Kriging: Properties and Implications on Simulation Experimental Design,” *ACM Transactions on Modeling and Computer Simulation*, 2024.
 - J. Zhang, X. Chen, and R. Wang, “Asymptotic Normality of Joint Metamodel-Based Sobol’ Index Estimators,” *Proceedings of the 2023 Winter Simulation Conference*, pp. 3705–3716, 2023.
 - J. Zhang and J. Chen, “Optimization of Multiple Distribution Center Location Problem Considering Flow Balance,” *Logistics Sci-Tech*, vol. 42, no. 10, pp. 15–17, 2019.
 - H. Zheng, J. Zhang, and B. Liu, “Integrated Continuous Berth Allocation and Quay Crane Scheduling under Tidal Influence at Container Terminals,” *Computer Integrated Manufacturing Systems*, vol. 24, no. 10, pp. 2599–2611, 2018.

- PROJECTS
- Multiple distribution center location for SAIC General Motors Company** 2019
Choose several distribution center locations using an improved K-means method
- Intelligent powertrain production planning system in SAIC General Motors Company** 2017 – 2018
Build a digital production planning system for an automobile company
- Intelligent analysis method of traffic saturation in ship alignment system** 2014

Design a method for measuring the traffic saturation in specific waters

AWARDS

Virginia Tech

- Graduate fellowship 2021

Shanghai Jiao Tong University

- First-class academic scholarship 2017 – 2020
- Silver award for the 3rd national logistics innovation competition 2018
- Outstanding graduates of Shanghai Jiao Tong University 2019

Dalian Maritime University

- National Merit Scholarship of China (top 1% GPA) 2014 – 2017

SKILLS

- Python, MATLAB, Git, L^AT_EX, Elementary Java/Shell/SQL